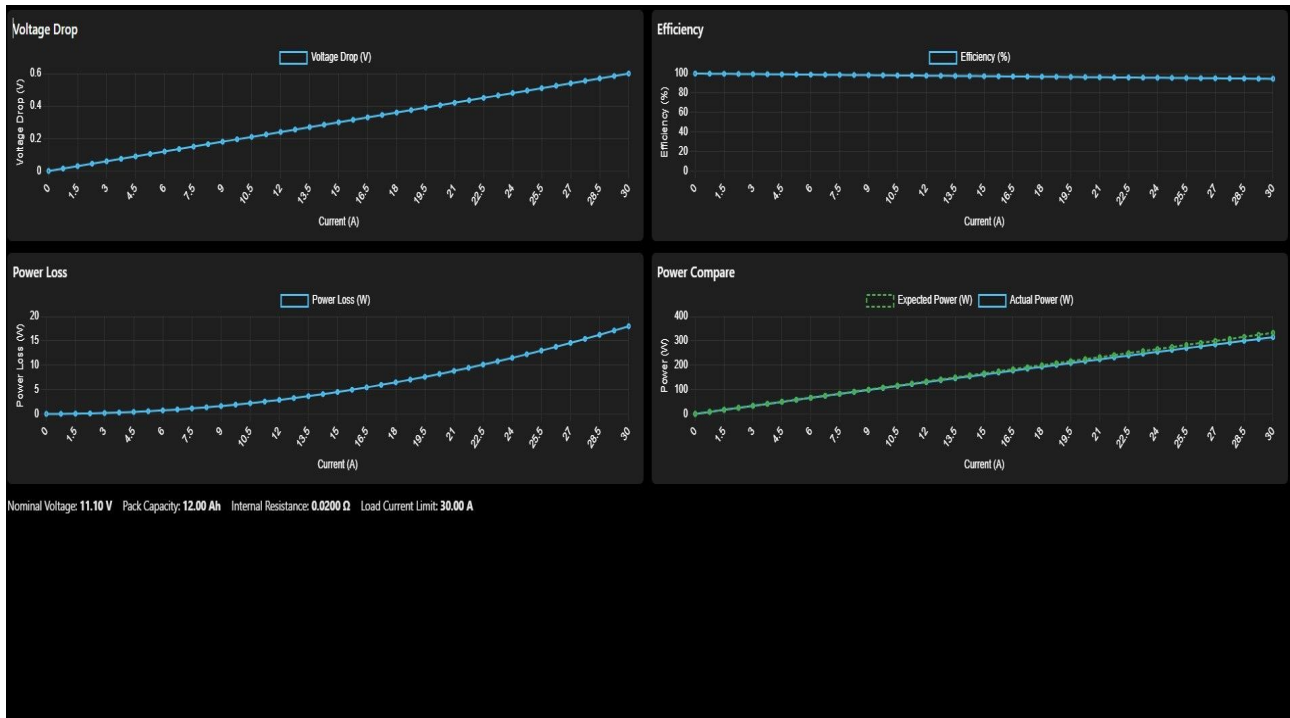


Battery Pack Design

3S6P Lithium-Ion Battery Pack — 12 Ah / 133.2 Wh



Design • Simulation • Hardware Implementation • Validation

This document presents the engineering design and implementation of a 3S6P 18650 lithium-ion battery pack, including electrical calculations, load-performance analysis, BMS integration, physical assembly, and validation considerations.

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Project Type: Hardware / Energy Storage / Battery Systems

1. Project Overview

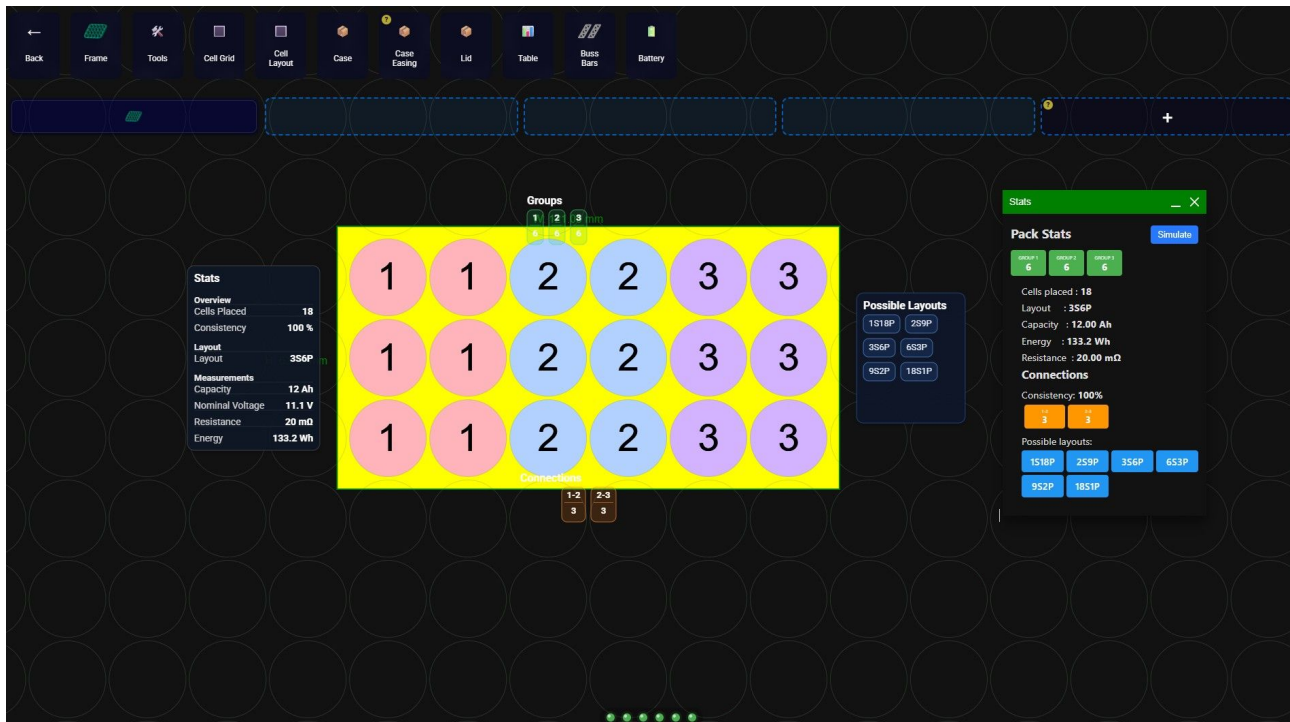
The objective of this project was to design, simulate, and physically assemble a compact lithium-ion battery pack using 18 cylindrical 18650 cells. The cells are configured as 3 series groups with 6 cells in parallel per group (3S6P). The design process combines analytical calculations with simulation-based performance evaluation before hardware realization.

Parameter	Specification
Cell type	18650 Li-ion
Cell nominal voltage	3.7 V
Cell capacity	2000 mAh (2.0 Ah)
Configuration	3S6P
Total cells	18
Nominal pack voltage	11.1 V
Maximum pack voltage	12.6 V
Pack capacity	12 Ah
Energy rating	133.2 Wh
Estimated pack resistance	20 mΩ
Analyzed load current	0–30 A
BMS	3S, 40 A protection/BMS

2. Battery Configuration

The pack uses three parallel groups connected in series. Each parallel group contains six cells. Parallel connection increases capacity and reduces effective resistance, while series connection increases the overall pack voltage.

Series groups	3
Cells per parallel group	6
Total cells	$3 \times 6 = 18$
Group nominal voltage	3.7 V
Pack nominal voltage	$3 \times 3.7 = 11.1 \text{ V}$
Group capacity	$6 \times 2 \text{ Ah} = 12 \text{ Ah}$
Pack capacity	12 Ah



3S6P cell-layout simulation used to validate the selected configuration.

3. Electrical Design Calculations

3.1 Pack Voltage

Nominal: $V_{nom} = 3 \times 3.7 = 11.1 \text{ V}$

Maximum: $V_{max} = 3 \times 4.2 = 12.6 \text{ V}$

3.2 Pack Capacity

For six identical 2 Ah cells connected in parallel:

$C_{pack} = 6 \times 2 \text{ Ah} = 12 \text{ Ah}$

3.3 Energy

$E = V_{nom} \times C = 11.1 \times 12 = 133.2 \text{ Wh}$

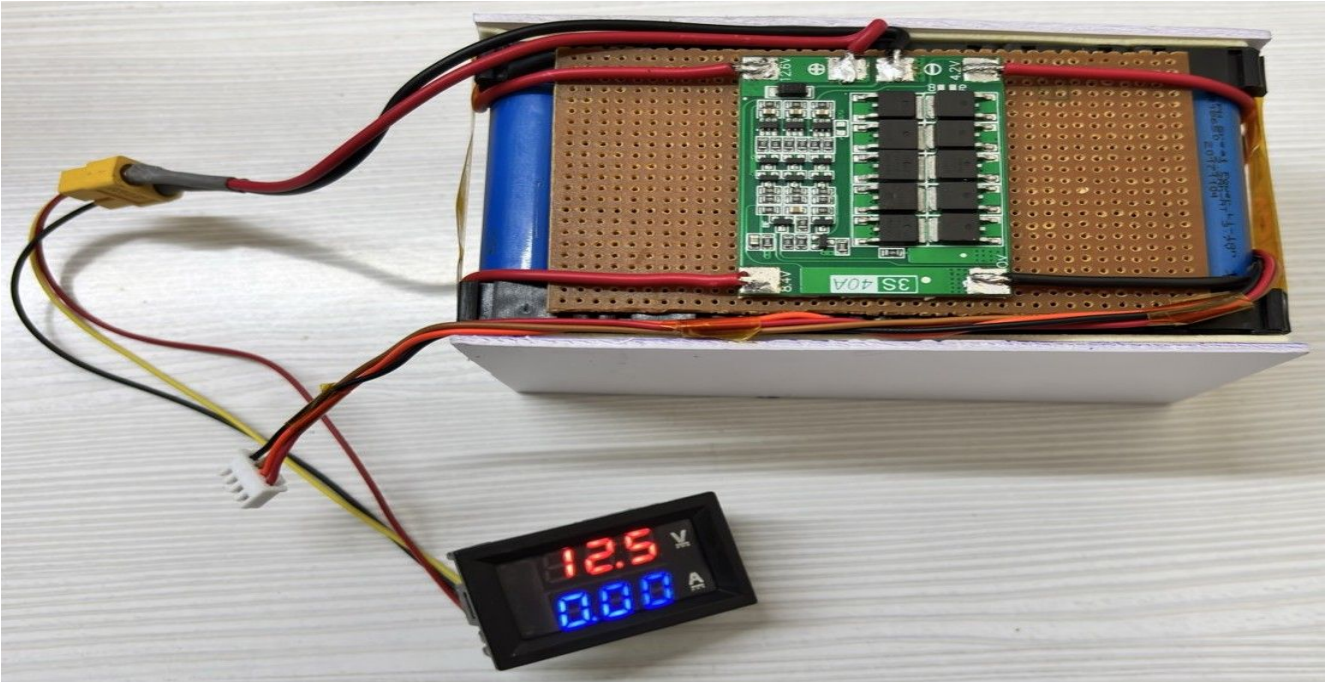
3.4 Internal Resistance Model

For six cells in parallel, the effective resistance of each group is reduced approximately by a factor of six. The three parallel groups are then connected in series. For the model used in the performance analysis, the resulting pack resistance is **20 mΩ (0.020 Ω)**.

Note: The 20 mΩ value is treated as the design/simulation model value. Actual pack resistance depends on cell condition, interconnect resistance, temperature, BMS MOSFETs, wiring, connectors, and measurement method.

4. Simulation & Performance Analysis

The pack model was evaluated over a 0–30 A load-current range. The analysis focused on voltage sag, resistive power loss, efficiency, and expected versus actual output power.



Simulation dashboard showing voltage drop, efficiency, power loss, and power comparison.

4.1 Voltage Drop

Using the simplified resistive model:

$$V_{\text{drop}} = I \times R$$

$$\text{At 30 A: } V_{\text{drop}} = 30 \times 0.020 = \mathbf{0.60 \text{ V}}$$

4.2 I²R Power Loss

$$P_{\text{loss}} = I^2 \times R$$

$$\text{At 30 A: } P_{\text{loss}} = 30^2 \times 0.020 = \mathbf{18 \text{ W}}$$

4.3 Efficiency

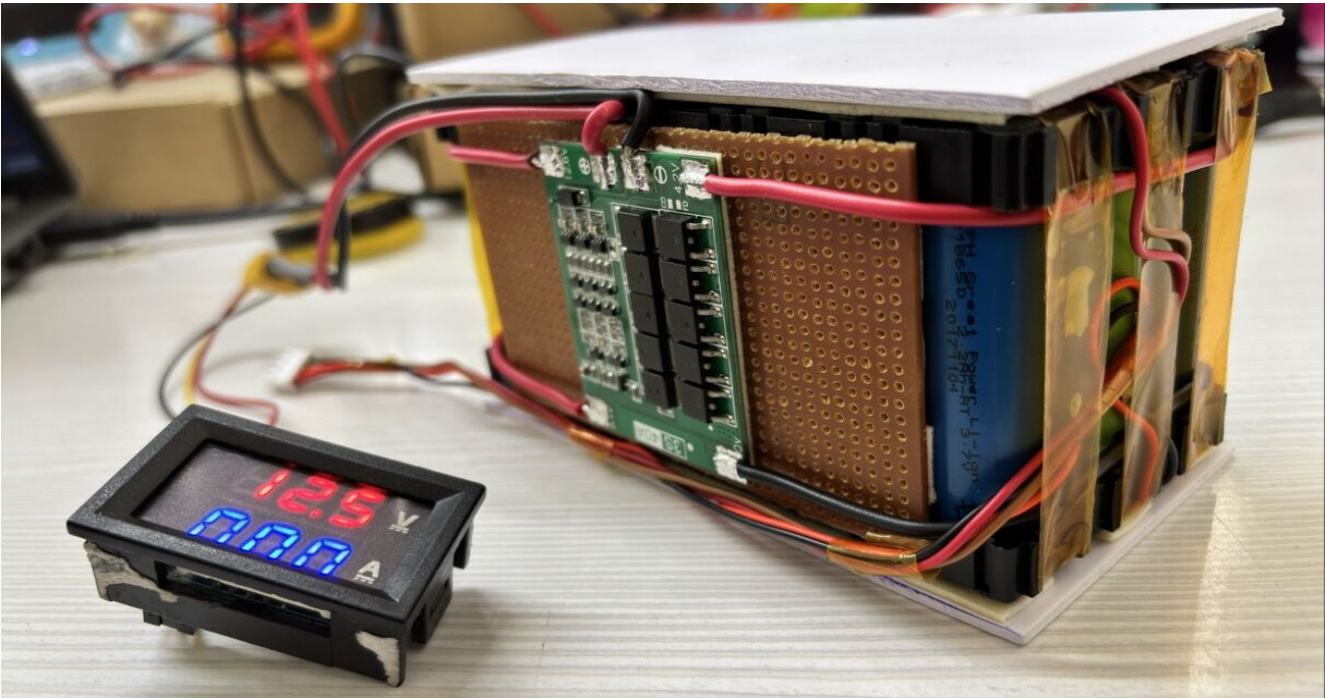
The modeled efficiency remains high at lower currents and decreases gradually as resistive losses increase. At 30 A, the modeled result is approximately **94–95%**.

Parameter	At 30 A
Nominal voltage	11.1 V
Modeled resistance	20 mΩ
Voltage drop	≈ 0.60 V
I ² R loss	≈ 18 W
Estimated efficiency	≈ 94–95%

Engineering interpretation: Voltage sag and I²R losses increase with load current. The analysis therefore provides an early indication of electrical and thermal stress before high-current physical testing.

5. Hardware Implementation

The physical pack was assembled using 18 18650 cells in a 3S6P configuration and integrated with a 3S 40 A BMS. The pack includes a main power connector and a balance connector for monitoring the individual series groups.



Physical pack assembly and BMS integration.

Hardware Components

Component	Specification	Qty
18650 Li-ion cells	3.7 V, 2000 mAh	18
BMS	3S, 40 A	1
Main connector	XT60	1
Balance connector	4-pin	1
Voltage/current meter	Panel DC V/A display	1
Mechanical structure	Custom pack frame/enclosure	1

6. BMS Protection & Battery Safety

The integrated 3S BMS is used as the battery protection and balancing subsystem. The intended protection functions include:

- Overcharge protection
- Over-discharge protection
- Over-current protection
- Short-circuit protection
- Passive balancing of series groups

Safety: Lithium-ion cells can present fire, thermal, and electrical hazards if incorrectly assembled, charged, discharged, shorted, or mechanically damaged. Properly matched cells, insulation, protection circuitry, charging control, wiring, mechanical restraint, and temperature monitoring are essential for safe operation.

7. Validation & Test Plan

The simulation results should be treated separately from experimentally measured results. The following validation sequence is recommended for the assembled pack:

Test	Purpose
Individual cell voltage check	Confirm cell condition and matching
Series-group voltage check	Verify 3S configuration
BMS connection check	Verify protection and balance wiring
No-load pack voltage	Confirm nominal/maximum operating range
Controlled load test	Measure voltage sag and output power
Temperature measurement	Assess thermal behavior under load
Capacity test	Measure usable Ah/Wh
Balance test	Verify series-group voltage equalization

8. Engineering Outcomes

- Applied series/parallel lithium-ion pack design principles.
- Calculated pack voltage, capacity, energy, internal resistance, voltage sag, and I^2R losses.
- Used simulation to evaluate performance across a 0–30 A load range.
- Integrated a 3S BMS for protection and passive balancing.
- Translated the electrical design into a physical battery pack.
- Established a workflow separating modeled performance from experimental validation.

9. Future Improvements

- Active cell balancing
- Individual cell-group voltage and temperature monitoring
- Coulomb counting and SOC estimation
- SOH estimation
- CAN-based BMS communication
- Battery data logging and visualization
- Controlled electronic-load characterization
- Thermal characterization and enclosure optimization
- Dedicated PCB-based BMS integration

10. Conclusion

The project demonstrates a complete hardware-oriented battery development workflow, from 3S6P configuration and electrical modeling through BMS integration and physical assembly. The 11.1 V nominal, 12 Ah pack provides approximately 133.2 Wh of nominal energy. The modeled 20 m Ω resistance predicts approximately 0.60 V voltage drop and 18 W resistive loss at 30 A. These results provide a baseline for subsequent controlled-load and thermal

testing of the physical pack.

Project Repository: 3S6P-Li-Ion-Battery-Pack-Design

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